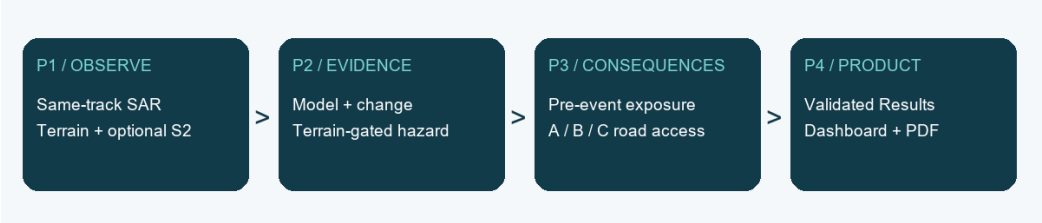
Cutoff / Flood exposure and access

Educational prototype - not an operational tool. Exposure is not confirmed damage.

Abstract

Cutoff is an educational prototype for rescue planning. Given an area and event time, it combines same-track radar change, a flood model, terrain and pre-event infrastructure to answer three questions: where did the event leave evidence, what was exposed, and who may be cut off? The product is a map dashboard and one-page situation report driven by a validated Results record.

System overview



Current delivery

P1-P3 algorithms and the P4 product are implemented. Pipeline/CLI, strict PDF, dashboard, feature CI, isolated-install checks, guarded evaluation and runtime tooling are delivered. Complete Trishuli north/south and Deir exposure/access integration is recorded in P4_LIVE_INTEGRATION.md. E4/freeze and human rehearsal remain explicit handoff checks.

Users and boundaries

Settlement-scale prioritisation and evidence review, rather than structural damage assessment. Every display carries the prototype disclaimer. Missing observations and unmapped access are represented as uncertainty.

Source: docs/plan/00_MASTER_PLAN.md; docs/plan/reports/P4.1-P4.7.md; P4_LIVE_INTEGRATION.md; PROJECT_AUDIT_2026-10-15.md

Methods / consistent observations

Acquisition and preprocessing

Choose a post-event pass strictly after the supplied UTC timestamp and up to three pre-event passes from the same relative orbit/direction. Require VV/VH and adequate footprint coverage. Bare dates mean end of day. Store linear sigma-zero, apply Lee Sigma and average pre-event scenes in linear units. Fast processing uses CDSE Sentinel Hub; full processing uses SNAP. Buffered Copernicus GLO-30 provides slope, HAND and drainage.

Model and evidence

The seven-channel U-Net uses pre/post VV/VH dB, their differences and slope, with the released no-slope model holding the slope channel at its training mean. Calibration/threshold selection uses validation events only. A scene-wide robust change scale identifies coherent decreases/increases in the drainage corridor. Terrain/S2 evidence adjusts confidence; shadow/layover and invalid data become no observation.

Exposure and access

Intersect hazard with strictly pre-event OSM. Split roads at valid intersections, retain keyed parallel edges, and compare baseline/post-closure travel time to mapped targets. Scenario A uses proximity to all hazard, B sustained high-confidence overlap, C bridge approaches. Every scenario closes bridges only on high-confidence approach evidence. Report connected/degraded/cut off/unknown separately and retain trail access. No operational closure is asserted.

Product integrity

Validate stage outputs on the shared UTM grid before consumers run. Optional optical errors continue; infrastructure errors publish a partial report; core acquisition errors preserve their diagnostic. The dashboard and PDF read Results statistics. OSM caches are isolated by snapshot/settings in P4.

Source: docs/plan/05_contracts.md; configs/default.yaml; docs/plan/reports/P1_HANDOFF.md; P2_HANDOFF.md; P3_HANDOFF.md

Held-out evaluation / global floods

E1 - measured flood-class performance

Model	Precision	Recall	F1	IoU
b0	0.720	0.640	0.678	0.512
b1_snunet_pre_post	0.849	0.768	0.807	0.676
b3_no_slope	0.813	0.819	0.816	0.689

Source: docs/results/raw/test_b0_b1_early_b3.json; docs/results/P2_kuro_results.md

E2 - final model by slope stratum

Slope bin	Flood support pixels	IoU	F1
0-5 deg	10887913	0.692	0.818
5-15 deg	164760	0.556	0.714
>=15 deg	8742	0.351	0.520

Source: same raw file, final model strata, ALL events

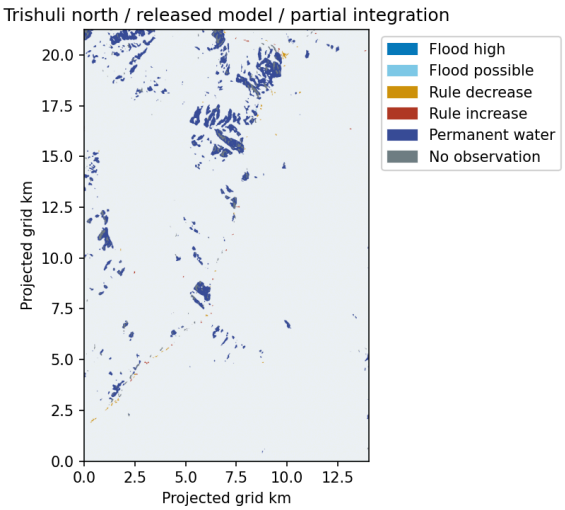
The official event-level split reduces within-event leakage. Training/calibration decisions were made before the Himalayan check. Steep-terrain support is much smaller than lowland support, so global scores do not establish Himalayan reliability. Model scores are calibrated global-flood scores, not verified local probabilities.

Other measured experiments

P1 documents same-track/cross-track and fast/full comparisons (E5/E3c). The fast/full stable VV median difference was 0.59 dB and the downstream hazard agreement was sensitive to change-scale differences. These are limitations to report, not reasons to tune on the check-only case.

Source: docs/results/E3c.md; docs/results/E5.md; docs/results/raw/val_e8.json

Case studies / integration and timing



Source: docs/report/evidence/c2-trishuli-north-model.json and its run hazard_class.tif; modified Copernicus Sentinel data 2026

E7 - actual pipeline measurements

Real run	Mode	Pipeline seconds	Peak RSS MB	State
c1-trishuli-north-b0	fast	64.78	764.75	partial
c2-trishuli-north-model	cached	56.96	1308.47	partial
c2-deir-ez-zor-model	cached	56.47	1325.57	partial

Source: docs/results/raw/E7.json; named real run.log files

Timings use warm satellite caches or cached mode, omit CLI startup, and stop with unavailable OSM/access. RSS includes child processes and excludes GPU VRAM. They do not substitute for a cold 10/20/40 km sweep. [TARGET] 20 x 20 km fast mode within ten minutes remains unverified.

Limitations / evidence still pending

E4 and E6

No real EMSR927 comparison has occurred. The check-only harness is complete and tested with synthetic polygons. After final review and rehearsal, freeze clean code/config/model and run it once. The optical observed-event layer represents mass movement, creating a definition mismatch with SAR water. Point-only reference buildings do not establish footprint recall. Human tile taxonomy remains necessary.

E6 tooling and algorithm tests exist. Dated 15 km OSM coverage and complete A/B/C integration are verified; the frozen real-run dropout table and external transport check remain pending. E6 bridge counts use original way counts from Results, not densified segments.

Scientific and operational limits

Overpasses can miss the peak; this is not early warning. Ten-metre pixels omit narrow roads/bridges. Shadow/layover are unobservable. Wet mud may resemble water; rough debris is statistical change. Surface DEM and gorge smoothing influence HAND/slope. Permanent-water scores are gated by terrain and clear optical evidence and excluded from hazard exposure. Historical case figures and runtimes predate this integration. OSM incompleteness can change access status; unknown means unmapped, not safe. Roads outside the observed AOI are assumed open.

Readiness limits

Competition permission for cached/fast processing and pretrained encoders is unconfirmed. Independent teammate fresh-clone and unprepared-AOI rehearsal remain pending. Submission evidence must retain these distinctions; targets and synthetic demonstrations are not real-world accuracy results.

Source: docs/results/E4.md; docs/results/E6.md; docs/results/E7.md; docs/plan/reports/organiser_answers.md; issues #25 and #31

Attribution / reproducibility / handoff

Required credit

Contains modified Copernicus Sentinel data 2026.

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Flood model trained on Kuro Siwo (Bountos et al., NeurIPS 2024 Datasets & Benchmarks), CC BY 4.0.

European Union, Copernicus Emergency Management Service data (check-only reference inventory/evaluation).

Reproduce the product

Follow README: clean environment, install dev/app/pdf/ml extras, create the labelled synthetic demo and start the local MapLibre dashboard with python app/server.py. The released model checksum is recorded in the P2 handoff. Feature branches and task reports document acceptance checks. Code licence: MIT; source data retain upstream licences.

Finish data-dependent evidence

1. Confirm access to the verified dated caches and working live credentials on the submission machine.
2. Rehearse complete Trishuli and a non-Trishuli AOI from a fresh checkout; validate contracts and dashboard.
3. Review final code/config/checkpoint, create frozen-v1, then run E4 once without retuning.
4. Run E6 and the cold-size E7 sweep; have a teammate rehearse a fresh clone and unprepared AOI.
5. Refresh this report, video and submission checklist from actual results before submitting.

Primary reproducibility sources

docs/plan/05_contracts.md; docs/results/P2_kuro_results.md and raw/test_b0_b1_early_b3.json; E3c.md; E5.md; E4.md; E6.md; E7.md and raw/E7.json; docs/plan/reports/P4.0-P4.9.md; docs/report/evidence/*.json; ATTRIBUTION.md.